Project Title: Advanced Automotive ECU: Performance Tuning and Safety Systems **Group Members:** Amr Waheed, Aly Hamdy, Zeyad Khaled, Abdelrahman Mohamed, Adham Amr

1. Motivation Our group chose to design an Advanced Automotive Electronic Control Unit (ECU) because it represents a cutting-edge, real-world application of digital logic in modern vehicle engineering. ECU hardware programming handles safety, fuel economy, and performance tuning in real time. By implementing four distinct vehicle subsystems on a single FPGA—specifically optimizing control signals for a 200V DC motor system—we can demonstrate how synchronous logic and rapid combinational circuits effectively manage critical mechanical functions. This project is highly engaging and relevant as it bridges core concepts of digital design with modern automotive performance engineering, and ensures the project is well scoped and feasible within the remaining time.

2. Approach

- **Overall Architecture:** The design features a unified top-level ECU module that processes multiple sensory inputs to control four independent automotive subsystems concurrently without blocking or data hazards. This approach clearly outlines how we plan to design and implement the system.
- **Main Modules:**
    1. *Parking Assist System:* A sequential counter that measures the width of an incoming ultrasonic echo_pulse. It triggers an alarm state if the pulse indicates a dangerously close obstacle.
    2. *Motor Controller:* A PWM generator that adjusts the duty cycle for a 200V motor driver based on a 2-bit throttle input and a selectable drive mode (Eco, Normal, Sport) using purely combinational decoding.
    3. *RPM Limiter:* A hardware comparator that continuously monitors engine RPM, instantly outputting an ignition_cut signal if the redline threshold is exceeded to protect the engine.
    4. *Active Aero System:* A speed-monitoring comparator that generates a spoiler_deploy signal to raise the rear aerodynamic wing when the vehicle's speed crosses a stability threshold.
- **Verification Strategy:** We will create a comprehensive Verilog Testbench to simulate real-world driving conditions, including varying echo pulses, throttle changes, and accelerating speed/RPM. We will analyze the resulting PWM waveforms and safety logic triggers in ModelSim/Vivado to ensure exact timing and structural stability before the final submission on Thursday, July 23.